Image tensors and channels

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One readable image-tensor and preprocessing record.

Trace values across axes,repair false reshapes and detect channel/double-scaling errors.

Keep one habit: W1–W3/W5 → selected new image → project pixel ledger → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Learned representations. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Initialisation and reproducibility. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Learning rate experiments. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- A small fictional RGB image has height 2 pixels, width 2 pixels and three **channel**s ordered red, green, blue. Store it as image[row][column][**channel**], abbreviated HWC.

  Use zero-based indices: the top row is 0, the left column is 0, and red, green and blue are **channel**s 0, 1 and 2. Each stored component is an integer from 0 to 255.

- The shape is 2 by 2 by 3. There are four pixels and twelve **channel** values. The blue component of the bottom-left pixel is image[1][0][2] = 255.

  A single **channel** is a plane of values across positions; it is not a separate photograph or a label.

- Shape needs a schema. A grayscale image might be stored as H by W or H by W by 1. An RGB image uses three **channel**s here; other formats can include additional **channel**s.

  Declare the format instead of assuming every three-dimensional array means the same thing.

Learned representations

> **My remembered idea, example and correction**

Initialisation and reproducibility

> **My remembered idea, example and correction**

Learning rate experiments

> **My remembered idea, example and correction**

Lesson reference for workbook practice

Position	RGB triplet	Meaning
Row 0 column 0	[255, 0, 0]	Red
Row 0 column 1	[0, 255, 0]	Green
Row 1 column 0	[0, 0, 255]	Blue
Row 1 column 1	[255, 255, 255]	White

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main2×2RGB, HWC[row][column][channel], channelsR/G/B0/1/2, zero-basedindices. Pixelsrow0 red[255, 0, 0]/green[0, 255, 0], row 1 blue[0, 0, 255]/white[255, 255, 255]. CHWplanesR[[255, 0], [0, 255]], G[[0, 255], [0, 255]], B[[0, 0], [255, 255]]. chw[c][r][k]=hwc[r][k][c]. Top-rightgreenHWC[0][1][1]=CHW[1][0][1]=255. Fourpixels, twelvevalues; five-imageNHWC5×2×2×3=60. Wrongflattenred[[255, 0], [0, 0]]differsbottomright.

  Divideonce:[255, 128, 0]→[1, 128/255, 0]; twice 255→1/255. Red/greensimplemeans 85 losecolour. Picture2×1 pixels[1, 4, 7]/[2, 5, 8], bottomgreen 5; three-imageNHWC3×2×1×3, batch[0][1][0][1]=5.

- For the lesson image, give image[1][0], image[1][0][2], the HWC shape, pixel count and number of **channel** values.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main2×2RGB, HWC[row][column][channel], channelsR/G/B0/1/2, zero-basedindices. Pixelsrow0 red[255, 0, 0]/green[0, 255, 0], row 1 blue[0, 0, 255]/white[255, 255, 255]. CHWplanesR[[255, 0], [0, 255]], G[[0, 255], [0, 255]], B[[0, 0], [255, 255]]. chw[c][r][k]=hwc[r][k][c]. Top-rightgreenHWC[0][1][1]=CHW[1][0][1]=255. Fourpixels, twelvevalues; five-imageNHWC5×2×2×3=60. Wrongflattenred[[255, 0], [0, 0]]differsbottomright.

  Divideonce:[255, 128, 0]→[1, 128/255, 0]; twice 255→1/255. Red/greensimplemeans 85 losecolour. Picture2×1 pixels[1, 4, 7]/[2, 5, 8], bottomgreen 5; three-imageNHWC3×2×1×3, batch[0][1][0][1]=5.

- A program flattens the lesson HWC array and groups the first four values into a 2 by 2 red plane. Give that incorrect plane and the correct plane. Identify a position where they disagree.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main2×2RGB, HWC[row][column][channel], channelsR/G/B0/1/2, zero-basedindices. Pixelsrow0 red[255, 0, 0]/green[0, 255, 0], row 1 blue[0, 0, 255]/white[255, 255, 255]. CHWplanesR[[255, 0], [0, 255]], G[[0, 255], [0, 255]], B[[0, 0], [255, 255]]. chw[c][r][k]=hwc[r][k][c]. Top-rightgreenHWC[0][1][1]=CHW[1][0][1]=255. Fourpixels, twelvevalues; five-imageNHWC5×2×2×3=60. Wrongflattenred[[255, 0], [0, 0]]differsbottomright.

  Divideonce:[255, 128, 0]→[1, 128/255, 0]; twice 255→1/255. Red/greensimplemeans 85 losecolour. Picture2×1 pixels[1, 4, 7]/[2, 5, 8], bottomgreen 5; three-imageNHWC3×2×1×3, batch[0][1][0][1]=5.

- Scale [255, 128, 0] by 255, keeping the middle value as a fraction. Then calculate the classroom RGB means of pure red and pure green. Which operation loses the distinction between those colours?

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main2×2RGB, HWC[row][column][channel], channelsR/G/B0/1/2, zero-basedindices. Pixelsrow0 red[255, 0, 0]/green[0, 255, 0], row 1 blue[0, 0, 255]/white[255, 255, 255]. CHWplanesR[[255, 0], [0, 255]], G[[0, 255], [0, 255]], B[[0, 0], [255, 255]]. chw[c][r][k]=hwc[r][k][c]. Top-rightgreenHWC[0][1][1]=CHW[1][0][1]=255. Fourpixels, twelvevalues; five-imageNHWC5×2×2×3=60. Wrongflattenred[[255, 0], [0, 0]]differsbottomright.

  Divideonce:[255, 128, 0]→[1, 128/255, 0]; twice 255→1/255. Red/greensimplemeans 85 losecolour. Picture2×1 pixels[1, 4, 7]/[2, 5, 8], bottomgreen 5; three-imageNHWC3×2×1×3, batch[0][1][0][1]=5.

- The lesson image is loaded with red and blue exchanged, then scaled twice. Give two expected checks that would expose these errors even though the shape is correct.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A one-row two-column RGB image has pixels [10, 20, 30] then [40, 50, 60]. State its HWC shape and all three CHW planes. Give the green value at row 0 column 1 in both layouts.

  A batch contains four copies: give its NHWC shape and total **channel**-value count.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Convert a new image independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Add a labelled image **tensor**s and **channel**s **evidence** sheet to Image-pattern Network Investigation. Show how you can read a pixel using declared row column and **channel** axes.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Audit an axis conversion for Image **tensor**s and **channel**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Use the worked case for Image **tensor**s and **channel**s. Proposed explanation: “Three **channel**s mean three images”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A one-row two-column RGB image has pixels [10, 20, 30] then [40, 50, 60]. State its HWC shape and all three CHW planes. Give the green value at row 0 column 1 in both layouts.

  A batch contains four copies: give its NHWC shape and total **channel**-value count.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Move colour values between **tensor** layouts without changing the pixels.
- Toy RGB image has 2 rows, 1 column. Top pixel [1, 4, 7]; bottom [2, 5, 8]. HWC means height, width, **channel**.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

- Where is the bottom green pixel in the first batched image? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-5/20

Two RGB pixels before layout conversion

	red	green	blue
top pixel	1	4	7
bottom pixel	2	5	8

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/20

Image tensors and channels

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Record HWC shape

Shape is [2, 1, 3]; top green is 4 and bottom green 5.

↓

2. Reorder to CHW

Shape [3, 2, 1]; red plane [[1],[2]], green [[4],[5]], blue [[7],[8]].

↓

3. Check same pixel

Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5.

CHANGE ONE THING

Add a batch containing three identical images in NHWC layout.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Move colour values between **tensor** layouts without changing the pixels.
- Which labelled step matches this detail? Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5.
- 1. Record HWC shape
- 2. Reorder to CHW
- 3. Check same pixel
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Check same pixel Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5. A **tensor** is an array with named dimensions.

HWC orders height, width and colour; CHW puts colour first. Rearranging the layout changes where values are stored, not the colour measurement itself. Trace a named pixel through both layouts using zero-based indices.

Adding a batch dimension counts images and does not add a new colour **channel**.

Changed-case question and optional hint

- Changed-case practice: Add a batch containing three identical images in NHWC layout. Which conclusion is supported by the supplied facts?
- 1. Batch shape is [3, 2, 1, 3].
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Batch shape is [3, 2, 1, 3]. The new first dimension counts images; it does not add a fourth colour **channel**.

Repair and retain the record

Review Learned representations (ai-5/19). Goal: Trace an **input** through a hidden representation and a decision rule. Short practice: Why can no deterministic head using only the sum classify both correctly?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Initialisation and reproducibility (ai-5/17). Goal: Trace how starting values affect a fixed **training** budget. Short practice: Does the matching final loss mean the parameter values are equal?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Learning rate experiments (ai-5/14). Goal: Run a controlled comparison of learning rates. Short practice: What should a fair rate comparison hold fixed?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | | |
> |---|---|
> | | |
> | | |
> | | |
> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Image tensors and channels: the case

AI ACADEMY / AI-5/20 · 01 · LOOK AT THE FACTS

Image tensors and channels: the case

The situation

Move colour values between tensor layouts without changing the pixels.

Supplied evidence

Toy RGB image has 2 rows, 1 column. Top pixel [1, 4, 7]; bottom [2, 5, 8]. HWC means height, width, channel.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain image tensors and channels using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

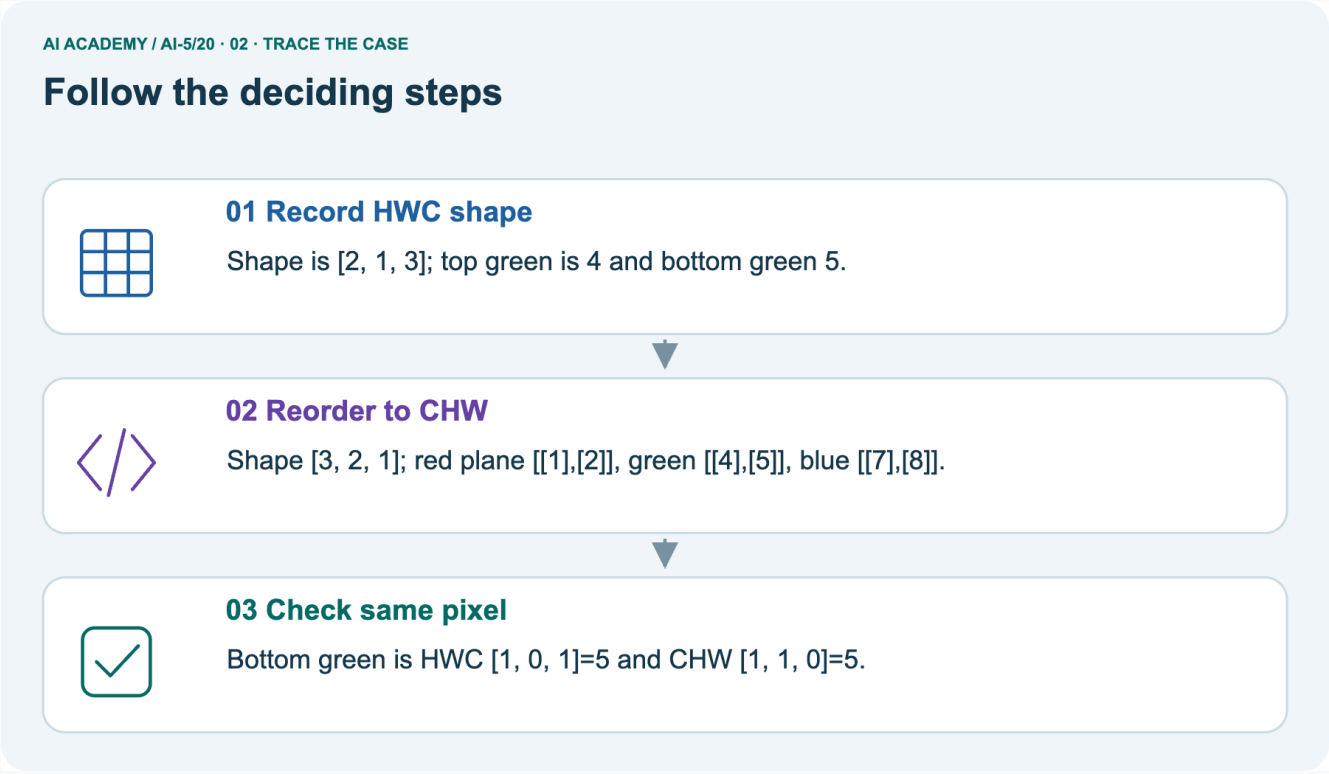
Move colour values between **tensor** layouts without changing the pixels. Supplied facts: Toy RGB image has 2 rows, 1 column. Top pixel [1, 4, 7]; bottom [2, 5, 8]. HWC means height, width, **channel**.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



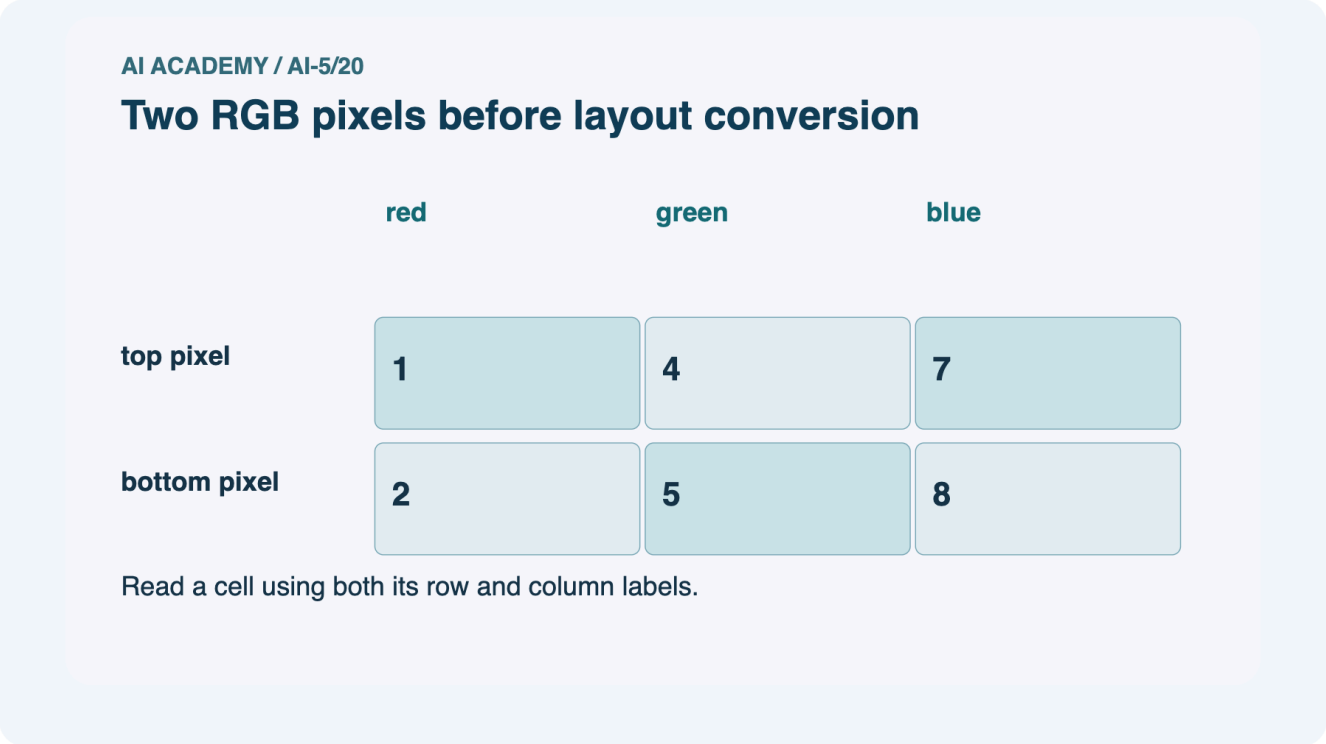
Follow the labelled picture with the matching explanation. Open the vector diagram

Record HWC shape: Shape is [2, 1, 3]; top green is 4 and bottom green 5. → Reorder to CHW: Shape [3, 2, 1]; red plane [[1],[2]], green [[4],[5]], blue [[7],[8]]. → Check same pixel: Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Two RGB pixels before layout conversion



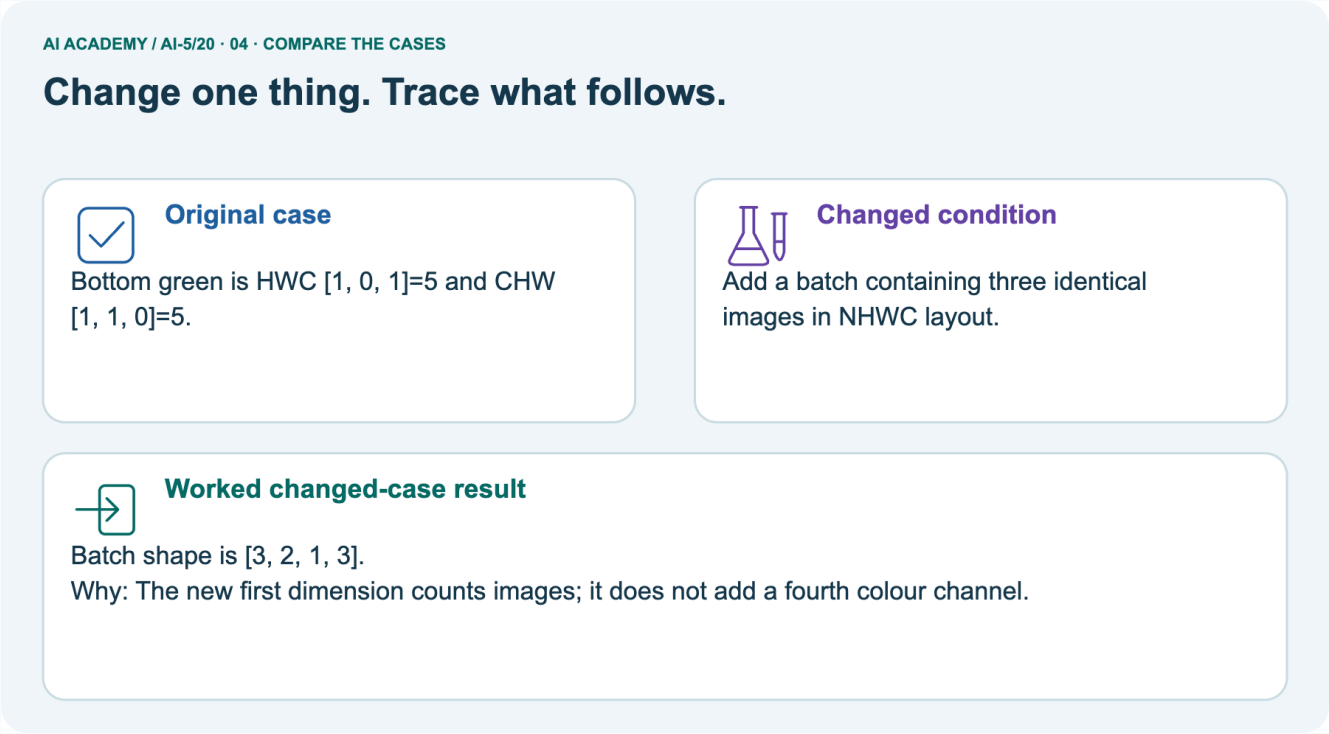
Follow the labelled picture with the matching explanation. Open the vector diagram

Two RGB pixels before layout conversion. top pixel: red = 1, green = 4, blue = 7. bottom pixel: red = 2, green = 5, blue = 8.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

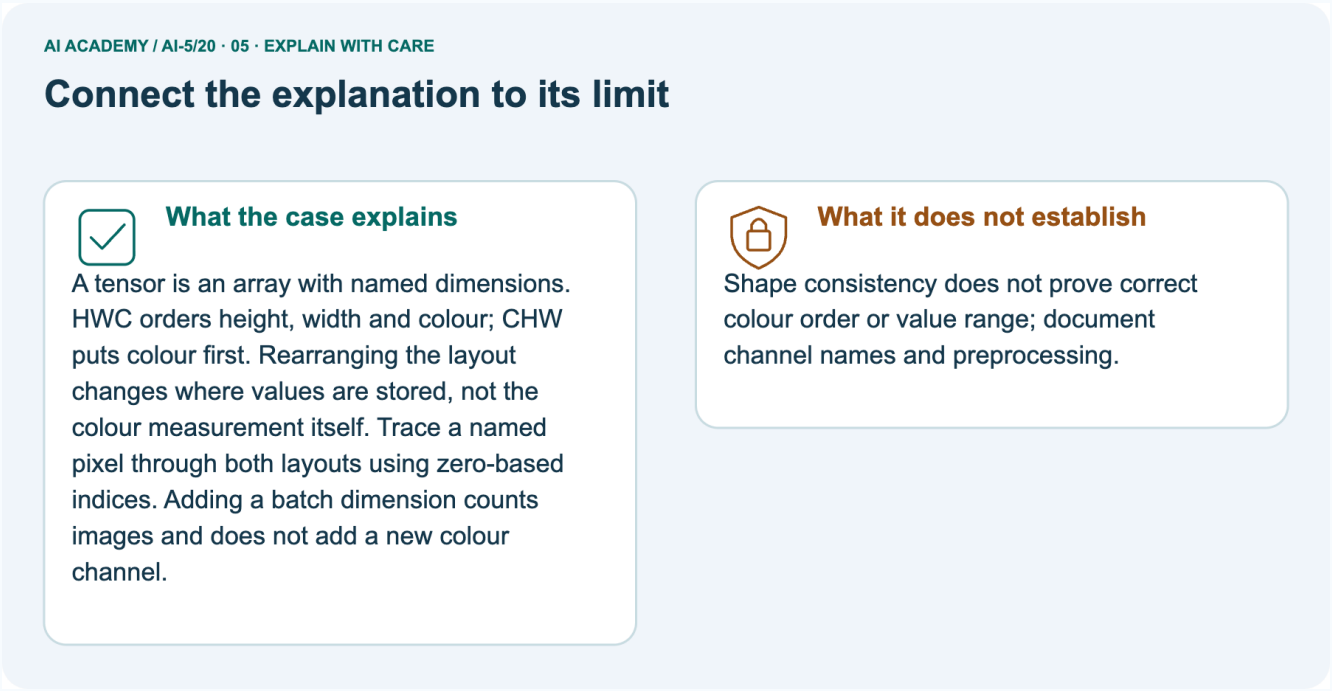
Original case: Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5. Changed condition: Add a batch containing three identical images in NHWC layout. Expected result: Batch shape is [3, 2, 1, 3].

Reason: The new first dimension counts images; it does not add a fourth colour **channel**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **tensor** is an array with named dimensions. HWC orders height, width and colour; CHW puts colour first. Rearranging the layout changes where values are stored, not the colour measurement itself.

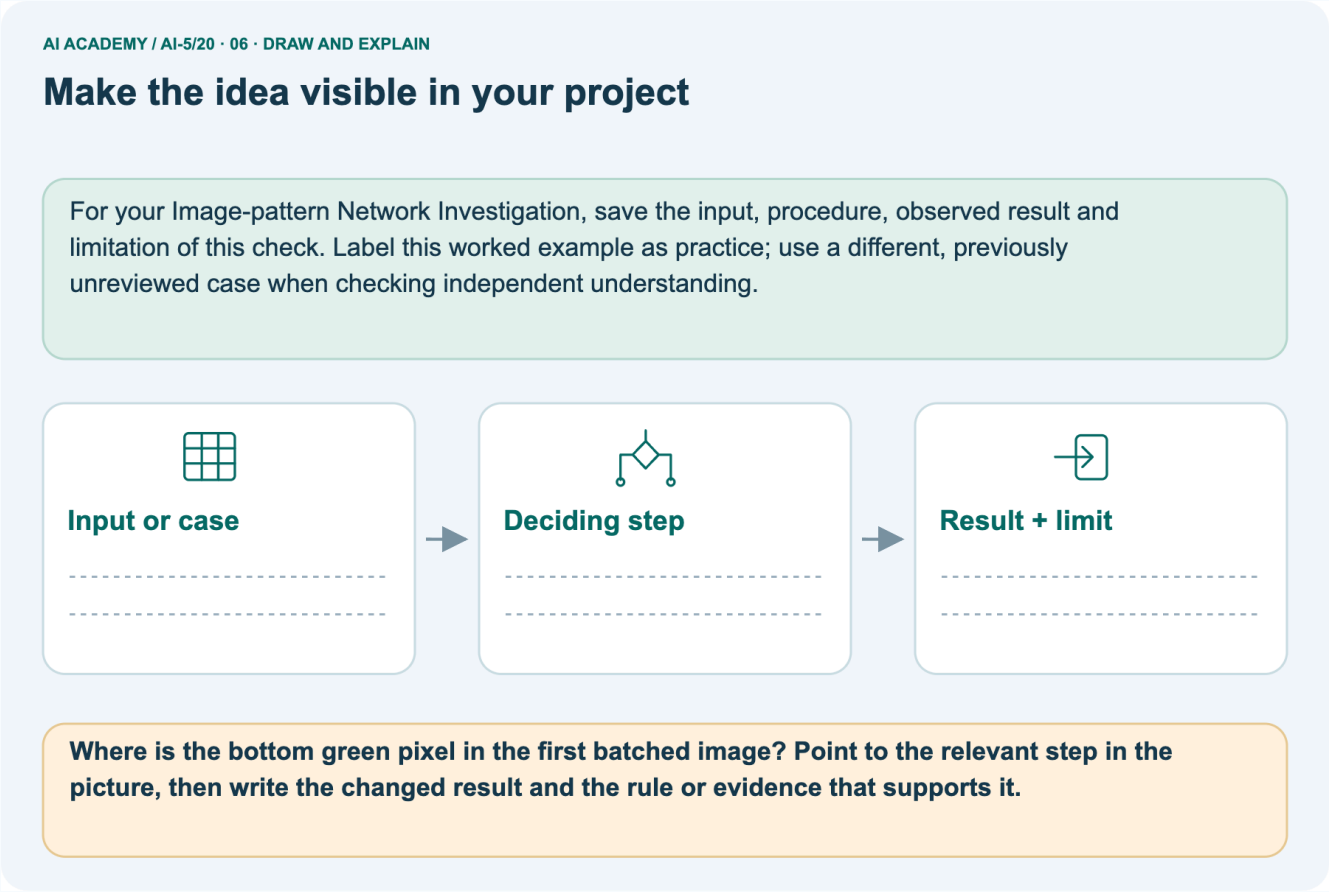
Trace a named pixel through both layouts using zero-based indices. Adding a batch dimension counts images and does not add a new colour **channel**.

Limit: Shape consistency does not prove correct colour order or value range; document **channel** names and pre**process**ing.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Where is the bottom green pixel in the first batched image? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Image tensors and channels: the case

02. Follow the deciding steps

03. Two RGB pixels before layout conversion

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.